

ON THE NINU SUPERMARKET POS SYSTEM

Entity Relationship Diagram (SQL Server)

1) ADMIN / OPERATIONS

role
PK id int
name nvarchar
desc nvarchar

role_permission
PK id int
FK role_id int
module nvarchar
action nvarchar
desc nvarchar

branch
PK id int
name nvarchar
address nvarchar
phone nvarchar
FK manager_id int
is_active bit

department
PK id int
FK branch_id int
name nvarchar
FK manager_id int

employee
PK id int
first_name nvarchar
last_name nvarchar
national_id nvarchar
email nvarchar
FK dep_id int
FK role_id int
FK branch_id int
salary decimal(18,2)
hire_date datetime
is_active bit

register
PK id int
FK branch_id int
register_code nvarchar
status nvarchar
is_active bit

shift
PK id int
FK register_id int
FK employee_id int
start_time datetime
end_time datetime
opening_balance decimal(18,2)
closing_balance decimal(18,2)

4) AUTHENTICATION

user_account
PK id int
FK employee_id int
username nvarchar
password_hash nvarchar
is_active bit
created_at datetime

2) CATALOG & SUPPLY

supplier
PK id int
sup_name nvarchar
contact_name nvarchar
phone nvarchar
email nvarchar
address nvarchar
tax_num nvarchar
pay_terms_days int
is_active bit

category
PK id int
FK parent_id int
cat_name nvarchar
is_active bit

product
PK id int
FK category_id int
FK supplier_id int
product_name nvarchar
barcode nvarchar
cost_price decimal(18,2)
sell_price decimal(18,2)
stock_qty int
reorder_level int
is_active bit

inventory_movement
PK id int
FK product_id int
movement_type nvarchar
qty int
movement_date datetime
reason nvarchar

discount
PK id int
disc_name nvarchar
type_d nvarchar
value_d decimal(18,2)
st_date datetime
end_date datetime
is_active bit

discount_product
PK id int
FK discount_id int
FK product_id int

purchase_order
PK id int
FK supplier_id int
FK employee_id int
order_date datetime
expected_date datetime
total_amount decimal(18,2)
status_po nvarchar

po_item
PK id int
FK po_id int
FK product_id int
qty int
cost_price decimal(18,2)
total_cost decimal(18,2)

LEGEND

- One
- > Crow's Foot (Many)
- Mandatory Relationship
- Optional Relationship

3) SALES & CUSTOMER

customer
PK id int
cus_name nvarchar
phone nvarchar
email nvarchar
loyalty_points int
is_active bit
registered_at datetime

loyalty_account
PK id int
FK customer_id int
card_number nvarchar
points_balance int
membership_level nvarchar
created_at datetime

points_transaction
PK id int
FK loyalty_id int
points_change int
transaction_type nvarchar
transaction_date datetime

sale
PK id int
FK customer_id int
FK employee_id int
FK register_id int
sale_date datetime
total_amount decimal(18,2)
tax_amount decimal(18,2)
discount_amount decimal(18,2)
final_amount decimal(18,2)

sale_item
PK id int
FK sale_id int
FK product_id int
qty int
unit_price decimal(18,2)
total_price decimal(18,2)

payment
PK id int
FK sale_id int
amount decimal(18,2)
pay_method nvarchar
pay_date datetime
status_p nvarchar

sale_return
PK id int
FK sale_id int
FK customer_id int
return_date datetime
total_return decimal(18,2)
reason nvarchar

sale_return_item
PK id int
FK sale_return_id int
FK product_id int
qty int
refund_amount decimal(18,2)